Aejandro De La Torre
LIS 440
Lecture: *wk11: Lecture-Demo: State Housing Data*
11/11/25

Code

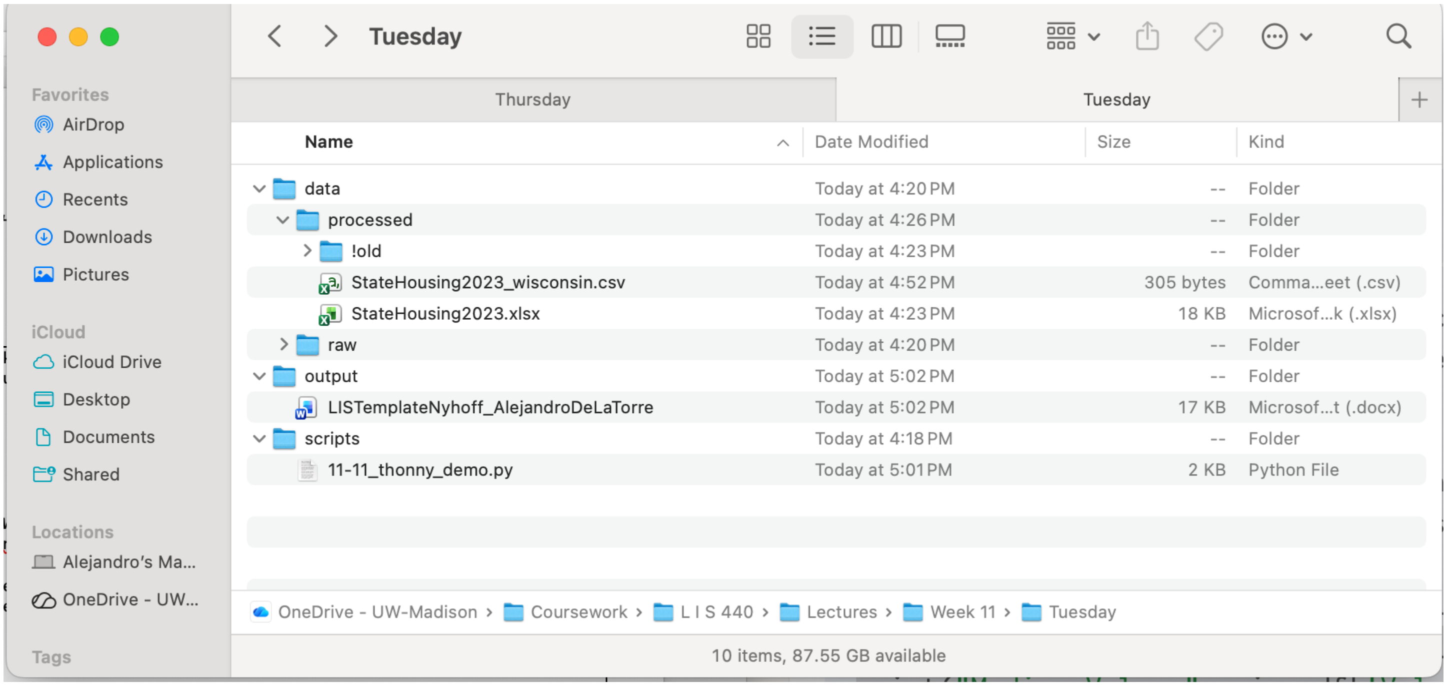
```
# Import the necessary packages
import pandas as pd # because 6 characters is to much
import os
from pathlib import Path # to set up paths easier

# Print version of pandas
print(pd.__version__)

# Set working directory (CWD) : root folder
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory: ",Path.cwd())

# Set CWD: Data
data = root / 'data'
# Set CWD: Data > Raw
data_raw = data / 'raw'
# Set CWD: Data > Processed
data_processed = root / 'data' / 'processed'
# Set CWD: Scripts
scripts = root / 'scripts'


# Read Student Data
wisc_data = data_processed / 'StateHousing2023_wisconsin.csv'

# Create dataframe of student data
wisc_df = pd.read_csv(wisc_data) # He labeled it wisconsinHousingDF but too long


# View first and last 3 entries
print("First 5 rows: \n", wisc_df.head(3)) # just show head
print("Last 5 rows: \n", wisc_df.tail(3)) # check if I am there

# Ask for shape of Function
print("Shape of data: \n", wisc_df.shape)
print("Columns:", wisc_df.columns )

# Get the number of rows and columns using .shape
num_rows, num_cols = wisc_df.shape
print(f"Total number of rows: {num_rows}")
print(f"Total number of columns: {num_cols}")
```

```
# Trying out summary stat funcs from
https://pandas.pydata.org/docs/getting_started/intro_tutorials/06_calculate_statistics.
html
print(wisc_df.describe())


# Following what prof does:
print("Last 2 rows: \n", wisc_df.tail(2))

valueColumnSeries = wisc_df["Value"]
print("Value Column: ", valueColumnSeries)

# Convert the Value column from String to Numeric
# 1. Remove the '$' and ',' characters
wisc_df['Value'] = wisc_df['Value'].str.replace(r'[$,]', '', regex=True)
# 2. Convert the cleaned column to a numeric type
wisc_df['Value'] = pd.to_numeric(wisc_df['Value']) # numeric to then be able to
calculate mean, etc
print(wisc_df.head(1)) # check if convert successfull

# Print the corrected col value entries
print("The Values column has ", valueColumnSeries.size, "entries.")

# Basic summary stats
print("Average Value (Mean):", wisc_df['Value'].mean())
print("Median Value:", wisc_df['Value'].median())
print("Mode Value(s):\n", wisc_df['Value'].mode())



print("End of program.")
```

Shell

```
>>> %Run 11-11_thonny_demo.py
2.3.3
Current Working Directory:  /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 11/Tuesday/scripts
First 5 rows:
     Month        Value
0  1/31/23  $268,736.68
1  2/28/23  $270,601.00
2  3/31/23  $273,370.26
Last 5 rows:
      Month        Value
9   10/31/23  $285,186.25
10  11/30/23  $285,404.52
11  12/31/23  $285,129.45
Shape of data:
 (12, 2)
Columns: Index(['Month', 'Value'], dtype='object')
```

```
Total number of rows: 12
Total number of columns: 2
        Month        Value
count       12          12
unique      12          12
top    1/31/23  $268,736.68
freq         1           1
Last 2 rows:
      Month        Value
10  11/30/23  $285,404.52
11  12/31/23  $285,129.45
Value Column:  0     $268,736.68
1     $270,601.00
2     $273,370.26
3     $276,247.66
4     $279,194.59
5     $281,146.74
6     $282,099.93
7     $283,201.35
8     $284,443.19
9     $285,186.25
10    $285,404.52
11    $285,129.45
Name: Value, dtype: object
    Month      Value
0  1/31/23  268736.68
The Values column has  12 entries.
Average Value (Mean): 279563.4683333333
Median Value: 281623.33499999996
Mode Value(s):
 0    268736.68
1     270601.00
2     273370.26
3     276247.66
4     279194.59
5     281146.74
6     282099.93
7     283201.35
8     284443.19
9     285129.45
10    285186.25
11    285404.52
Name: Value, dtype: float64
End of program.
>>>
```


Tuesday
Favorites
AirDrop
Applications
Recents
Downloads
Pictures
iCloud
iCloud Drive
Desktop
Documents
Shared
Locations
Alejandro's Ma...
OneDrive - UW...
Tags
Thursday
Tuesday
Name
Date Modified
Size
Kind
data Today at 4:20 PM -- Folder
processed Today at 4:26 PM -- Folder
!old Today at 4:23 PM -- Folder
StateHousing2023_wisconsin.csv Today at 4:52 PM 305 bytes Comma...eet (.csv)
StateHousing2023.xlsx Today at 4:23 PM 18 KB Microsof...k (.xlsx)
raw Today at 4:20 PM -- Folder
output Today at 5:02 PM -- Folder
LISTemplateNyhoff_AlejandroDeLaTorre Today at 5:02 PM 17 KB Microsof...t (.docx)
scripts Today at 4:18 PM -- Folder
11-11_thonny_demo.py Today at 5:01 PM 2 KB Python File
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